Practice Questions

Lectures (8-12)

Question no.1
Definition: Echelon form, reduced echelon form.

Question no.2
If $A=\begin{bmatrix}2 & -2 & 1\\ 2 & 6 & 5\\ 0 & 0 & 1\end{bmatrix}$, $I=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ then find A^2-3I.
Where I is the identity matrix of order 3*3
Answer:
$\begin{bmatrix}-3 & -16 & -7\\ 16 & 29 & 37\\ 0 & 0 & -2\end{bmatrix}$
Question no.3
Find the matrix of cofactors of the given matrix
$C_{ij}=\begin{bmatrix}-2 & -1 & 1\\ 5 & -3 & 5\\ 4 & 6 & 2\end{bmatrix}$

Answer:
$C_{ij}=\begin{bmatrix}-36 & 10 & 42\\ 8 & -8 & 8\\ -2 & 5 & 11\end{bmatrix}$

Question no4
Given matrices A and B such that
$A=\begin{bmatrix}-4 & -2\\ 9 & 8\end{bmatrix}$ $B=\begin{bmatrix}0 & -5\\ 5 & 8\end{bmatrix}$
Evaluate A-3B.

Answer: $\begin{bmatrix}-4 & 13\\ -6 & -16\end{bmatrix}$

Question no.5

Check whether the matrix $A=\begin{pmatrix} 4 & 8 & 6 \\ 11 & -6 & 5 \\ 9 & -4 & 7 \end{pmatrix}$ and $B=\begin{pmatrix} 5 & 6 & 8 \\ 4 & 7 & 9 \end{pmatrix}$ can be multiplied?

Justify your answer.

Solution:
order of A=3*3
Order of B=2*3
Inner dimensions do not match so this multiplication is not possible.

Question no.6

Find the inverse of A=$\begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix}$

Answer: $\begin{bmatrix} -4/2 & 2/2 \\ -3/2 & 1/2 \end{bmatrix}$

Question no.7

Find the determinant of the following matrix

$A=\begin{bmatrix} -3 & 4 & 9 \\ 6 & -1 & -5 \\ 2 & 8 & 7 \end{bmatrix}$

Answer: 143